Title: Rethinking Concept Drift Adaptation: A Dimensionality Reduction Methods Perspective

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Abstract:

Concept drift adaptation remains a critical challenge for data warehousing, particularly in urban mobility analysis. This paper proposes a framework built on dimensionality reduction methods that explicitly accounts for reproducibility of analytical workflows. We evaluate the framework on a ten-year retail transaction log and compare it against 3 baseline methods. Experimental results show that the proposed approach improves silhouette coefficient by 18.8% while keeping forecast MAPE within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of data warehousing has changed how researchers and practitioners approach concept drift adaptation, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, dimensionality reduction methods have attracted sustained attention as a promising route toward solving concept drift adaptation in realistic settings such as urban mobility analysis. Despite considerable progress, existing solutions for concept drift adaptation tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in forecast MAPE frequently come at the expense of silhouette coefficient, forcing designers to accept trade-offs that are poorly understood.

Our work is motivated by the growing demand for urban mobility analysis, where reliability and transparency requirements are especially strict. We make three contributions: a unified formulation of concept drift adaptation, a practical design built on dimensionality reduction methods, and an empirical evaluation on a ten-year retail transaction log. Understanding reproducibility of analytical workflows is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

OConnor et al. (2013) introduced an early framework based on dimensionality reduction methods, reporting a 5.1% improvement in forecast MAPE over previous baselines. Subsequent work by Ahmed et al. (2015) extended this line of research to urban mobility analysis, although the evaluation was restricted to small-scale settings. A survey by Fernandez et al. (2012) categorised existing approaches to concept drift adaptation into three families and highlighted the lack of standard benchmarks. Researchers working on data warehousing have explored probabilistic topic models as an alternative, with Mukherjee et al. (2012) noting that the approach scales poorly beyond 32 components. Johansson et al. (2014) analysed reproducibility of analytical workflows and showed that purely heuristic solutions cannot guarantee consistent silhouette coefficient.

More recently, Hoffmann et al. (2013) combined dimensionality reduction methods with probabilistic topic models, achieving a 8.3% reduction in overhead on a ten-year retail transaction log. In contrast to these efforts, our work treats concept drift adaptation and reproducibility of analytical workflows jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to concept drift adaptation as a persistent obstacle for urban mobility analysis. Despite this body of literature, no prior study has

systematically compared dimensionality reduction methods and probabilistic topic models under identical experimental conditions.

3. Methodology

We formalise concept drift adaptation as a constrained optimisation task in which the objective is to maximise silhouette coefficient while keeping RMSE within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on dimensionality reduction methods, and a validation layer that incorporates probabilistic topic models. All experiments were conducted on a ten-year retail transaction log, which provides 1200 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 30 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. Our implementation is written in Python and C++, and was executed on a workstation with 64 GB of memory and a single mid-range accelerator. We compare against 3 baseline methods, including a conventional approach built on probabilistic topic models and two recent variants reported in the literature. The proposed design improves silhouette coefficient by 18.8% relative to the strongest baseline and reduces silhouette coefficient by 9.3%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the probabilistic topic models component reduces silhouette coefficient by approximately 4.0%, confirming its contribution. A sensitivity analysis over 8 parameter settings shows that performance remains stable, with variation in RMSE below 13.5%. Threats to validity include the limited diversity of a ten-year retail transaction log and the possibility that deployment in urban mobility analysis introduces conditions not captured in our setup.

4. Conclusion

This paper investigated concept drift adaptation through the lens of dimensionality reduction methods, and demonstrated that a carefully designed pipeline can improve silhouette coefficient by 18.8% on a ten-year retail transaction log. The findings suggest that data warehousing will continue to benefit from closer integration of dimensionality reduction methods and probabilistic topic models, particularly in urban mobility analysis. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of reproducibility of analytical workflows. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for concept drift adaptation.